



MECH MAINT Department
SIMHADRI

Note No :1



Ref:SMTPP/MECH MAINT/2023-24/LMI/499476

Date:20/09/2023(12:49:33)

Subject:Approval for Revision of LMI

The Location Management Instruction (LMI) of Performance monitoring of CW system i.e LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 revision is done as per revision of TC document.

In this LMI Changes are made in monitoring sheet of CW/CT/Condenser performance.ACC condenser performance sheet is included in the check list.

Approving authority for LMI is HOP.

Submitted for kind approval.

Submitted by:

पुष्पिता बेहेरा - वरिष्ठ प्रबंधक

PUSPITA BEHERA - 101583 (SENIOR MANAGER ,MECH MAINT) ,Simhadri
(PUSPITABEHERA@NTPC.CO.IN, 9425534585)

On:20/09/2023(12:49:33) **Ref:**SMTPP/MECH MAINT/2023-24/LMI/499476

Note No:2

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:21/09/2023(09:11:48) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:3

Submitted by:

पुष्पिता बेहेरा - वरिष्ठ प्रबंधक

PUSPITA BEHERA - 101583 (SENIOR MANAGER ,MECH MAINT)
(PUSPITABEHERA@NTPC.CO.IN, 9425534585) ,Simhadri

On:21/09/2023(09:58:15) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:4

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:21/09/2023(10:00:59) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:5

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:21/09/2023(17:46:08) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:6

LMI is reviewed and found in line with OGN.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:22/09/2023(09:18:29) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:7

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:23/09/2023(17:19:12) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:8

Please review.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,OPERATION)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:23/09/2023(18:25:51) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:9

LMI Reviewed w.r.t to OGN and LMI of other stations and observations are recorded as below.

Page No 7 of LMI:

Point C:

- ii) Acid cleaning to be explored only after cleaning and thereafter assessment of tubes by Eddy Current method(presently in practice). Responsibility for the same may shared by TMD/EEMG & Chemistry.
- iii) Presently Mechanical cleaning is being done by High pressure Jet followed by Conco Bullet Cleaning.

Page No 8 of LMI

Point e: Following the Helium leak test protocol is to be jointly signed with EEMG Operation Chemistry and TMD. Responsibility for the same lies with EEMG/Chemistry/Operation/TMD.

The above changes may be incorporated.

Put up for further review.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)
(RRSATPATHY@NTPC.CO.IN, 9437021584) ,Simhadri

On:04/10/2023(19:46:11) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:10

Changes proposed in Note Page-5 to be incorporated.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,OPERATION)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:05/10/2023(09:44:02) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:11

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:05/10/2023(11:40:20) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:12

Required changes done. Please review.

Submitted by:

पुष्पिता बेहेरा - वरिष्ठ प्रबंधक

PUSPITA BEHERA - 101583 (SENIOR MANAGER ,MECH MAINT)
(PUSPITABEHERA@NTPC.CO.IN, 9425534585) ,Simhadri

On:05/10/2023(12:24:06) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:13

Forwarded

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:05/10/2023(12:30:38) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:14

Submitted by:

धर्मेंद्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:06/10/2023(10:03:30) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:15

Critically review from the role of EMD in the LMI in LT/MUW/HT areas

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:07/10/2023(11:06:15) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:16

HT motor maintenance will be in EMD-HT scope

Submitted by:

सतीश कुमार सिंह - उप महाप्रबंधक

Satish Kumar Singh - 008418 (DY. GENERAL MANAGER ,ELECT MAINT)
(SATISHKUMARSINGH@NTPC.CO.IN, 8004940879) ,Simhadri

On:09/10/2023(09:40:47) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:17

Maintenance of Hydraulic Pump Motor and Motorized Butterfly Valve are in EMD-LT scope.

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

KRISHNA BHESETTY - 009044 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(BKRISHNA@NTPC.CO.IN, 9493546044) ,Simhadri

On:09/10/2023(17:54:57) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:18

Submitted by:

मेडा नाग राजा - उप महाप्रबंधक

Meda Naga Raja - 008404 (DY. GENERAL MANAGER ,ELECT MAINT)
(MEDANAGARAJA@NTPC.CO.IN, 9437578452) ,Simhadri

On:16/10/2023(11:08:59) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:19

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:16/10/2023(15:53:28) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:20

Reviewed

Submitted by:

दीपशिखा वर्मा - उप महाप्रबंधक (रसायन)

Deepshikha Verma - 007773 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(DEEPSHIKHAVERMA@NTPC.CO.IN, 9650995244) ,Simhadri

On:20/10/2023(10:29:30) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:21

Reviewed and forwarded for approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

On:20/10/2023(13:21:00) **Ref:**SMTTP/MECH MAINT/2023-24/LMI/499476

Note No:22

Please check the second page (p no. 2) of the revised LMI (copy attached in Annexures). If, no LMI review is to be done by GM(Mtnc.) or GM(O&M) then why do you route it so ?

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:27/10/2023(18:44:29) **Ref:**SMTPP/MECH MAINT/2023-24/LMI/499476

Note No:23

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:27/10/2023(18:47:56) **Ref:**SMTPP/MECH MAINT/2023-24/LMI/499476

Note No:24

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:28/10/2023(11:12:21) **Ref:**SMTPP/MECH MAINT/2023-24/LMI/499476

Note No:25

The matter was discussed with EEMG and the same file is being routed for according approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:28/10/2023(11:29:23) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

Note No:26

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:04/11/2023(16:58:04) **Ref:**SMTTPP/MECH MAINT/2023-24/LMI/499476

NTPC Ltd

LOCATION MANAGEMENT INSTRUCTION

Performance Monitoring of CW System

LMI/SMSTPS/EEMG/OGN/OPS/PERF/008

Rev:02 Sept:2023

Approved for Implementation by: **HOP (SMSTPS)**

Enquiries to: **HOD (EEMG)**

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

LMI APPROVAL DOCUMENT

1	LMI NAME	Performance Monitoring of CW System
2	LMI No.	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev: 02
3	Reference Document	COS-ISO-00-OGN/OPS/PERF/008/ Rev:02 August:2023
4	Superseded Document	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008/ Rev:01 Dec:2021
5	Prepared By	Name: Puspita Behera Designation: Sr. Manager (EEMG)
6	Checked By	Name: Ajay Gupta Designation: DGM (EEMG)
7	Reviewed By	Name: Uma Shanker Verma Designation: AGM (EEMG) Name: Malay Kanti Jana Designation: AGM (OPERATION) Name: Shambhu Kumar Suman Designation: AGM (TMD) Name: Dharmendra Kumar Tomar Designation: AGM (EMD) Name: Piyush Srivastava Designation: AGM (CHEMISTRY) Name: G. Hari Narayana Rao Designation: AGM (C&I)
8	Approved By	Name: Sanjay Kumar Sinha Designation: GGM (Simhadri)

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

INDEX

SL NO	CONTENT	PAGE NO
1	Introduction	4
2	Superseded Document	4
3	Records of Revision	4
4	Objective	4
5	Electro-chlorination	4
6	CW & CT forebay Chlorination	5
7	CW Pump	5
8	Travelling Water Screen (TWS) in Sea Water Pump House	6
9	Debris Filter	6
10	Condenser Performance	7
11	Cooling Tower Performance	8
12	Water Balance	8
13	Review	8
14	Annexure -I	9
15	Annexure -II	10

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

1.0 INTRODUCTION

Cooling water system is one of the key systems in designing the power station and plays a vital role during operation of power station. In the steam condenser more than 60 % of the total heat input to the plant is finally dissipated as waste heat. This heat is carried away by the circulating water in the power system.

Closed cooling water system is adopted at Simhadri where the circulating water from condenser is cooled in a cooling tower and water is re-circulated back in the system. The losses due to the blow down (to prevent building up of salt concentration), evaporation in cooling tower is made by means of make up to the system.

2.0 SUPERSEDED DOCUMENTS

This supersedes LMI/OGN/OPS/PERF/008: Performance Monitoring of CW system Rev -01 issued in Dec:2021.

3.0 RECORDS OF REVISION

LMI has been revised due to the TC document revision. Changes are made in monitoring sheet of CW / CT / condenser performance. ACC condenser performance sheet is included in the check list.

4.0 OBJECTIVE

To enhance the performance of equipment such as travelling water screens, CW pumps, Condenser, Cooling towers, CW chlorination plant etc. during operation of CW system.

- a. Identification of areas in which improvement can be made in order to have a reliable, efficient and economic generation.
- b. To check and monitor the consumption of water on regular basis (raw water & clarified water) for various purposes i.e. equipment cooling, ash water, CW circulation through condenser, cooling tower etc. in order to improve efficiency and optimize make up water consumption.

5.0 ELECTRO-CHLORINATION

The electro-chlorination system is provided at the sea water pump house intake basin for the control of organic growth in the sea water well, pipelines etc. the system is designed to generate sodium hypochlorite as biocide by the electrolysis of sea water. The equipment consists of bipolar cells, electrolyser to generate 20 Kg/hr of sodium hypochlorite per stream.

Dozing is done directly at the intake well at the effluent discharge pipeline. The residual chlorine at the pump discharge is maintained at 1 ppm and 0.5 ppm at the CW fore-bay.

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

Continuous dosing is done at 20 Kg/hr and shock dosing at 40 kg/hr at every 8 hrs.

- a. Ensure healthiness of the rectifiers, dosing pumps & booster pumps.

Responsibility: EM / MM / C&I / EEMG

- b. Ensure operation of electro-chlorination for normal dosing and shock dosing.

Responsibility: Operation / Chemistry

- c. Check protection and interlocks once in a year.

Responsibility: Operation / C&I

6.0 CW & CT FOREBAY CHLORINATION

- a. Chlorination plant and chlorine absorption system are to be kept available in a healthy condition.

Responsibility: Chemistry / MM / C&I / EM

- b. Chlorine dosing at 20-30 kg/hr. continuously is carried out in addition 50-60 kg/hr chlorine is added as shock dosing for 3-4 hours in a day depending on the season. Residual chlorin at cooling tower basin is maintained around 1 ppm.

Responsibility: chemistry

- c. Chlorine line leakages shall be identified and arrested at earliest.

Responsibility: Chemistry / MM / C&I

7.0 CW PUMP

- a. Check the performance of CW pumps as per Annexure-I.

Responsibility: Operation

- b. Check CW pump discharge valve is in good operating condition. Refer LMI/OGN/OPS/MECH/008.

Responsibility: MM / EM / C&I

- c. Check the cooling water system for CW pump thrust bearing.

Responsibility: MM / Operation

- d. Ensure that bearings temperature do not exceed limit.

Responsibility: Operation

- e. Ensure seal water pumps availability.

Responsibility: Operation

- f. Check healthiness of interlocks and protections.

Responsibility: Operation / C&I

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

8.0 TRAVELLING WATER SCREEN IN SEA WATER PUMP HOUSE

Following points are to be monitored /checked.

- a. Healthiness of baskets in screen and nozzles.
Responsibility: Operation / MM
- b. See the chain link and its pin clearances. If it is more, then the same may be replaced.
Responsibility: MM
- c. Type & quantity of debris coming in screen.
Responsibility: MM
- d. Is there any carry over of debris towards CW pumps suction?
Responsibility: MM
- e. Level difference (DP) across the TWS.
Responsibility: MM
- f. Cleaning frequency of the filter in backwash system.
Responsibility: MM
- g. Debris outlet channel is to be checked for any blockage.
Responsibility: Operation
- h. TWS running hour sin a shift.
Responsibility: MM / Operation
- i. All spray nozzles are in position and clean.
Responsibility: MM
- j. Availability of spare parts.
Responsibility: MM

9.0 DEBRIS FILTER

Debris such as shells, sticks, plastics etc. are found to be common cause of chocking of condensers. Screening at the suction of CW pumps with the help of trash rack can only prevent entry of debris of larger size. Again, there is no screening of debris within the CW system itself. similarly, chlorination can kill mussels but not prevent the entry of shells into the condenser. Debris filter acts as the secondary screening very close to the water box.

When the debris enters the condenser tubes the CW velocity in those tubes increases. The same may be enough for the critical shear stress to be exceeded that leads to erosion-corrosion, a rapid form of attack. As the protective oxide film is removed the attack on tube material goes on un-impended.

Where extremely large number of tubes are clogged by debris the CW velocity in other tubes also increases leading to erosion-corrosion.

- a. If the DP across debris filter is in the increasing trend check whether the sequence of operation of debris filter is taking place.
Responsibility: Operation
- b. Check the functioning of rotating arm, flushing motor during overhaul.
Responsibility: MM / EM

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

10.0 CONDENSER PERFORMANCE

Factors that affect condenser performance are as follows.

a. High CW inlet temperature

Loss due to high CW inlet temperature occurs on account of high ambient WBT or poor performance of cooling tower. If cooling tower is performing satisfactorily, this loss is acceptable.

b. Low CW flow

In this case CW flow may be restricted due to fouled condenser tubes or chocking of water-box with foreign material or partial opening of condenser inlet or outlet valves. Water-box / tube cleaning to be done one by one by isolating one condenser pass at a time while unit running under part load or both passes cleaning to be done when condenser is out of service.

c. High Air ingress in Condenser /ACC

This is an entirely preventable loss and steps to be taken to locate and stop any air ingress into the system under vacuum as soon as condition is detected. The resistance to heat transfer of a layer of air 1.0 mm thick is equal to that of a slab of copper 16.5 mm thick. Many times, both vacuum pumps required to be run on account of this.

d. Dirty Tubes

Operationally little can be done to eliminate the cause of the loss, as tubes must be cleaned when condenser is out of service. However, as soon as loss is detected it should be ascertained that chlorine injection into the system is satisfactory. Station chemistry to comply with CW treatment guidelines of COS chemistry. All efforts to be made to prevent foreign material ingress into the system.

In order to optimize condenser performance following guidelines are to be taken care of:

- a.** Performance of condenser is to be analyzed as per annexure -II.

Responsibility: EEMG

- b.** Condenser tube cleaning last done (date to be recorded for frequency).

Responsibility: TMD

- c.** Method of cleaning condenser tubes.

- i.** On load tube cleaning (COLTCS to be kept in service)

Responsibility: Operation

- ii.** Acid cleaning is to be done only after cleaning and there after assessment of tubes by Eddy current method.

Responsibility: TMD / EEMG / CHEMISTRY

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

iii. Mechanical cleaning by high pressure jet followed by conco bullet cleaning.

Responsibility: Chemistry / TMD

d. Flood test last done(date), protocol to be made attending defects. No of tubes plugged to be recorded.

Responsibility: Operation / TMD / EEMG / Chemistry

e. Helium leak test done (date)- if increase in back pressure occur due to air ingress, plugging of air ingress points to be done at the earliest. Protocol to be made after attending defects and improvements in condenser back pressure to be recorded and jointly signed by TMD, Operation, EEMG & Chemistry.

Responsibility: TMD / Operation / EEMG / Chemistry

11.0 COOLING TOWER PERFORMANCE

Please refer document – Operation Guidance Note: Thermal performance of cooling towers, document no. COS-ISO-00-OGN/OPS/PERF/006

12.0 WATER BALANCE

Please refer document – Operation Guidance Note: thermal performance of cooling towers, document no. COS-ISO-00-OGN/OPS/PERF/006

13.0 REVIEW

The Head of project will be responsible for reviewing this document on a 2 yearly basis or as necessary.

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

ANNEXURE – I**LOG SHEET FOR CW/CT PUMP**

Station/Unit:

Date: -

Time: -

SI. NO.	PARAMETER	DESIGN	ACTUAL
1	Sump level (m)		
2	Discharge Pressure		
3	Motor Current		
4	Power Consumption		
5	Flow-Pitot/Annubar		
6	Vibration		
7	Bearing Temperature		
8	Impeller: Corro-coating: Date		
9	Last duct painting done: Date		
10	Healthiness of TWS		

Flow is calculated from characteristic curve with respect to available head.

- a. If the flow of the pump is less than the rated flow, the pump clearances may be checked during the pump overhaul.
- b. If the power consumption is more than the designed power, then the motor bearing and pump bearings may be checked during overhaul.

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

ANNEXURE – II**A. CONDENSER PERFORMANCE TEST**

Station/Unit:

Date:

Time:

Atmospheric Pressure:

SL. no	Parameter (Co. A)	Derivation (Co. B)	Unit	Test value
1	Condenser back pressure		mmHg (Abs)	
2	Corresponding saturation steam temp		deg. C	
3	Average CW inlet Temp. (Avg of LHS/RHS)		deg. C	
4	CW outlet Temp 'A'			
	'B'			
	'C'			
	'D'			
5	Avg. of CW outlet temp		deg. C	
6	Avg. of Exhaust Steam temp		deg. C	
7	Avg. of condensate temp		deg. C	
8	Avg. of air suction temp.		deg. C	
9	Position of CW outlet valves		%	
10	Design Condenser back pressure		mmHg (Abs)	
11	Design CW Temp rise		deg. C	
12	Design TTD		deg. C	
13	Back pressure due to CW inlet temp. (Saturation pr. Corresponding col. B temp)	(3) +(11) +(12)	mmHg (Abs)	
14	Back pressure due to CW flow (Saturation pr. corresponding to col. B temp)	(5) + (12)	mmHg (Abs)	
15	Back pr. variation due to CW inlet temp.	(13) - (10)	mmHg (Abs)	
16	Back Pr. variation due to CW flow	(14) - (13)	mmHg (Abs)	
17	Back Pr. variation due to air/dirty tube	(1) – (14)	mmHg (Abs)	

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023

18	Total condenser pressure variation	(1) – (13)	mmHg (Abs)	
19	No. of CW pumps in service	02		
20	No. of Vacuum pumps/ejectors in service	01		
21	Vacuum pump rotameter reading		Kg/hr. LPM	
22	COLTCS Running hours			
23	No. of balls in circulation	>90	%	
24	Ball Loss in 1000 hrs. of operation	<5%	%	
25	Condensate Dissolve Oxygen		PPM	
25	Condensate ACC/ PH/ Residual Chlorine			

B. ACC SYSTEM LOG SHEET (ROUTINE CHECK LIST)

SL no.	Parameter	Unit	Design value	Running parameter
1	Unit Load	MW		
2	Condenser Vacuum	mmHg		
3	LPT exhaust steam temperature	Deg.c		
4	Ambient air temperature	Deg.c		
5	ACC inlet and outlet air temperature	Deg.c		
6	Condensate temperature at CEP suction	Deg.c		
7	Condensate dissolved oxygen	PPB		
8	Nos. of fans running (Fast/Slow)	Nos.		
9	Last water washing	Date		

TITLE	Performance Monitoring of CW System
REF NO	LMI/SMSTPS/EEMG/OGN/OPS/PERF/008 Rev:02 Sept 2023